

How to Set up a Rework Station

No matter how precise our manufacturing process has become, manual rework of boards still exists. Also, rework is an integrated part of engineering in electronics and is considered so important that it is one of the first things taught in engineering schools. And this is also true that there is nothing like a complete rework station for an engineer. But you need to be little careful whilst setting up a rework station to get the maximum output of your work. Here is what you need to look at



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There are two most important things that you should consider whilst setting up a rework station. The first is that you must take all precautionary measures for electrostatic discharge (ESD)-sensitive components at the station. The second is that you should have all the correct tools for different jobs to make your rework more efficient.

ESD safety measures

Electronics is becoming smaller and faster but it is also becoming more sensitive towards ESD. The rework station needs to be specifically designed to minimise the effect of ESD, especially when various studies around the world have revealed that 60-90 per cent of defective devices are damaged

due to ESD, and 70 per cent of these failures can be attributed to damage caused by ungrounded workers. So it becomes really important that you take ESD-control systems seriously, or otherwise, the losses can be astonishingly high.

A basic ESD control rule is to ground all conductors including workers at the rework station. Grounding works very efficiently in ESD-control systems and reliably removes ESD to ground. For such a grounding system, it is important that the electrical wiring system of your lab is correct. All electrical outlets in your lab need to be evaluated for correct wiring of live, neutral and ground wires.

A wrist strap is an effective method for grounding the workers. The Electrostatic Discharge Association's standard ANSI/ESD S1.1-2006 defines

a wrist strap as an assembled device consisting of a wrist cuff and ground cord that provides electrical connection of a person's skin to the ground. The standard document completely describes the parameters for evaluation, acceptance and functional testing of wrist straps. Whilst the document describes the whole set of mechanical and electrical parameters over which a wrist strap needs to be evaluated and accepted, the most important parameter amongst all is the wrist strap continuity and resistance, which should be 1 mega-ohm ± 20 per cent, for acceptance. The document also suggests the testing procedure for the same. Whilst you are buying grounding materials, do check if they comply with the above-mentioned standard and specifications.



Fig. 1: ESD wrist strap

Now to ground rest of the conductors, including equipment lying on the workstations, ESD mats should be used which cover the complete bench top. Best practice is to use metal grounding hardware snaps and ground wires connecting the work surface mat to the common point ground. ANSI/ESD S4.1 standard defines test methods for evaluating and selecting work surface materials, new work surface installations and previously installed work surfaces.